

Tanveer

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About

BTech graduate in Biotechnology from Haldia Institute of Technology, India, worked at the intersection of computation and biology. My research focuses on with designing biosensing and diagnostic tools using Arduino, optical systems, and sensors. I am interested in interdisciplinary research combining biotechnology, bioinformatics, biophysical instrumentation, and molecular diagnostics, and seek to integrate artificial intelligence and machine learning approaches into experimental and analytical frameworks. Seeking a doctoral position at a reputable research organization to further pursue these interests and contribute to innovative solutions in life sciences.

Education

Haldia Institute of Technology, India

December 2021 – July 2025

BTech in Biotechnology

- GPA: 8.19/10.0 ([Link](#))
- **Some Relevant Coursework:** Biochem, Molbio, Microbio, Immunology etc.

Devasthaly Vidyapeeth, Uttar Pradesh, India

2019 – 2020

Senior School Certificate Examination

- Percentage: 71.4%
- **Relevant Coursework:** Physics, Chemistry, Mathematics, Computer Science

Research Experiences

Multi-Modal Neural Network for Tissue Classification Using CNN and Morphological Feature Fusion (IISER-Kolkata)

January 2026 - July 2026

Guide: Dr. Dipjyoti Das, Associate Professor

- Designed and implemented an end-to-end tissue image classification pipeline using convolutional neural networks for automated representation learning from microscopy images. Developed a parallel segmentation and feature-extraction workflow using multi-Otsu thresholding, watershed labeling, and Delaunay-based neighborhood graph construction to compute morphological and topological descriptors of cell organization. Combined CNN-based predictions with engineered geometric features in a fusion framework to improve classification performance and generalization.

Terahertz Spectroscopy of Glucose (IISER-Kolkata)

January 2025 - July 2025

Guide: Prof. Kamaraju Natarajan

- Prepared samples with varying glucose concentrations embedded in polystyrene pellets and performed terahertz time-domain measurements to extract their optical properties. The study aimed to determine the minimum detectable glucose concentration and evaluate the potential of terahertz spectroscopy as a non-invasive analytical tool for biosensing applications.

Presentation: [🔗](#)

Report: [🔗](#)

Design and Interfacing of stepper motor rotator for 1 inch and 2 inch polarization rotator (IISER-Kolkata)

June 2024 - July 2024

Guide: Prof. Kamaraju Natarajan

- Designed and implemented a wireless-controlled stepper motor rotator for 1-inch and 2-inch polarization rotators.
- Developed a computer application to control and monitor the rotator's operation and environmental conditions.

Additional work

- Created an AC current detection system with real-time alarming functionality.
- Analyzed and prototyped a gear-based mechanism to enhance torque for larger rotators.

Presentation: [🔗](#)

Additional Projects

- Portfolio Website**

Completed

 - Developed a responsive portfolio website using Next.js, Tailwind CSS, and Framer Motion. Implemented modern web design principles, optimized for performance and SEO, and hosted on Vercel.
Github Link: [Link](#)
- Computer Quiz App**

Ongoing

 - Developed a quiz app for PC using Java and NET Beans IDE. Implemented Oracle DBMS to store data. (Not yet uploaded on GitHub)

Skills

- Programming Languages:** Python, C++, C, Java, JavaScript, NextJS, Perl, SQL, Arduino, ESP-32, Linux, $\text{\LaTeX}$
- Software Tools & Frameworks:** Git, MATLAB, AutoDesk Fusion, PyTorch
- Bioinformatics Tools:** Sequence alignment, database management, BLAST, FASTA
- Spoken Languages:** English, Hindi

Couses and seminars attended

- Quantum Information and Quantum Technology**

Cert. [Link](#)

An International Conference, hosted by IISER Kolkata in 2023.
- NPTEL: The Joy of Computing using Python**

Cert. [Link](#)

Basics of Python programming language and a few libraries.

Extra-Curricular Activities

- Coding:** Hacker rank profile

Link [Link](#)
- Painting and sketching:** 3rd position in a college sketch competition.

Cert. [Link](#)
- Speed Cubing:** 3rd position in a cubing event at college tech fest.

Cert. [Link](#)
- Football:** Played in Intra-College Football Tournament.